

1.6 Electrochemistry

Question Paper

Course	CIEA Level Chemistry
Section	1. Physical Chemistry
Topic	1.6 Electrochemistry
Difficulty	Medium

Time allowed: 80

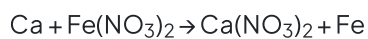
Score: /64

Percentage: /100

Question 1a

This question is about redox reactions.

When calcium reacts with $\text{Fe}(\text{NO}_3)_2$, a redox reaction occurs.



i)
Explain in terms of electrons, which species has been oxidised and which has been reduced.

[2]

ii)
Use oxidation numbers to show which species acts as the reducing agent.

[2]

[4 marks]**Question 1b**

What is the full systematic name for $\text{Fe}(\text{NO}_3)_2$?

[1 mark]

Question 1c

Another redox reaction occurs between cobalt oxide, Co_2O_3 , and hydrochloric acid, HCl .



i)

Explain, in terms of oxidation numbers, what has been reduced.

[2]

ii)

A student describes this reaction as disproportionation. Explain, using oxidation numbers, whether the student is correct or not.

[3]

[5 marks]

Question 1d

Magnesium oxide, MgO , reacts with nitric acid, HNO_3 , when heated.



i)

Identify the oxidation numbers of

Mg in MgO

N in HNO_3

Mg in $\text{Mg}(\text{NO}_3)_2$

N in $\text{Mg}(\text{NO}_3)_2$

[2]

ii)

Explain whether the reaction between magnesium oxide and nitric acid is a redox reaction.

[1]

[3 marks]

Question 2a

Under suitable conditions, SCl_2 reacts with water to produce a yellow precipitate of sulfur and a solution **A**. Solution **A** contains a mixture of $\text{SO}_2(\text{aq})$ and compound **B**.

What is the oxidation number of sulfur in SCl_2 ?

[1 mark]

Question 2b

i)
Deduce how the oxidation number of sulfur changes during the reaction of SCl_2 with water.

[2]

ii)
What type of reaction is this? Justify your answer.

[2]

[4 marks]**Question 2c**

i)
Suggest the identity of compound **B**.

[1]

ii)
Construct an equation for the reaction between SCl_2 and water.

[1]

[2 marks]

Question 2d

i)

What would you observe when each of the following reagents is added to separate samples of solution **A**?

$\text{AgNO}_3(\text{aq})$

$\text{K}_2\text{Cr}_2\text{O}_7(\text{aq}) / \text{H}^+(\text{aq})$

[2]

ii)

Acidified KMnO_4 would also react with solution **A**.

What is the systematic name of KMnO_4 ?

[1]

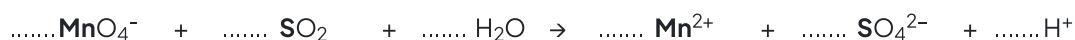
[3 marks]

Question 3a

This question looks at different redox reactions.

For each of the equations below, write the oxidation number of the element printed in **bold** underneath its symbol, and balance the equation by adding appropriate numbers before each species.

i)



oxidation numbers:

[3]

ii)



oxidation numbers:

[3]

[6 marks]

Question 3b

What would you **see** happening in the reaction shown in (a)(ii)?

[2 marks]

Question 3c

Zinc will displace copper from copper sulfate solution.

i)

Write the balanced symbol equation for the reaction.

[1]

ii)

Explain, using oxidation numbers, why this is a redox reaction.

[2]

[3 marks]

Question 3d

Use a half equation to explain what species is acting as the oxidising reagent in the reaction in (c).

[3 marks]

Question 4a

Household bleach contains NaClO as its active ingredient.

NaClO can be produced from the reaction between chlorine and water.

i)
State the conditions required for NaClO to be produced from chlorine and sodium hydroxide.

[1]

ii)
Give the systematic name of NaClO.

[1]

iii)
When chlorine and sodium hydroxide react under different conditions, sodium chlorate(V) is produced.
Give the formula for sodium chlorate(V).

[1]

[3 marks]

Question 4b

The balanced symbol equation showing the reaction of chlorine and sodium hydroxide to form NaClO is shown below.



i)

Give the ionic equation for this reaction.

[1]

ii)

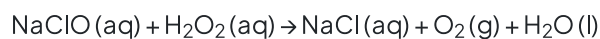
Explain, using oxidation numbers, why this is an example of a disproportionation reaction.

[3]

[4 marks]

Question 4c

The concentration of NaClO in bleach can be found by reacting it with hydrogen peroxide, H_2O_2 .



Explain, using oxidation numbers, why chlorine has been reduced in this reaction.

[1 mark]

Question 4d

A student reacted an excess of aqueous hydrogen peroxide to 10.0 cm^3 of a sample of bleach. They collected 108 cm^3 of oxygen gas.

i)

How many moles of oxygen gas were collected? Assume that under the conditions used, the molar gas volume is $24.0\text{ dm}^3\text{ mol}^{-1}$.

Show your working.

[1]

ii)

How many moles of NaClO were in the 10.0 cm^3 sample of bleach?

Show your working.

[1]

iii)

What was the concentration of NaClO, in mol dm^{-3} , of NaClO in the sample of bleach?

Show your working.

[1]

iv)

The label on the bottle of bleach gives the concentration of NaClO in g per 100 cm^3 .

Use your answer from part (iii) to deduce the concentration of NaClO in the bleach in g per 100 cm^3 .

Show your working.

[2]

[5 marks]

Question 5a

This question is about the redox reactions of halide ions.

Chlorine can displace iodine from potassium iodide.

i)

Write the balanced equation for this reaction.

[1]

ii)

Write the ionic equation for this reaction.

[1]

[2 marks]

Question 5b

Using oxidation numbers, explain how iodide ions act as a reducing agent in this reaction.

[2 marks]

Question 5c

i)

Chlorine also displaces bromine from potassium bromide but not as readily as it can displace iodine from potassium iodide.

Explain how you can use these results to determine which is the stronger reducing agent, bromide ions or iodide ions.

[2]

ii)

Explain why the halide ions show this trend in reducing power.

[4]

[6 marks]

Question 5d

Iodide ions can also act as a reducing agent to reduce Fe^{3+} ions to Fe^{2+} .

i)

Write the ionic equation for this reaction.

[2]

ii)

Explain, in terms of electrons, how Fe^{3+} ions act as an oxidising agent.

[2]

[4 marks]

Model Answers: Medium

1a

a)

i) The species that have been oxidised and reduced are:

- Ca / calcium has been oxidised as it has lost electrons; [1 mark]
- Fe / Fe²⁺ / iron has been reduced as it has gained electrons; [1 mark]

ii) The species that acts as the reducing agent is:

- Ca / calcium; [1 mark]
- As it reduces Fe / Fe²⁺ / iron from +2 to 0

OR

Its oxidation number increases from 0 to +2; [1 mark]

[Total: 4 marks]

- Calcium loses electrons:
 - $\text{Ca} \rightarrow \text{Ca}^{2+} + 2\text{e}^-$
- Fe²⁺ gains electrons:
 - $\text{Fe}^{2+} + 2\text{e}^- \rightarrow \text{Fe}$
- **Remember:** OIL RIG – Oxidation Is Loss, Reduction Is Gain of electrons
- So Ca is oxidised and Fe²⁺ is reduced
- A reducing agent is one which reduces another species, so is itself oxidised
- You can show that Ca is the reducing agent by either showing, using oxidation numbers, that it is oxidised or that Fe²⁺ is reduced

1b

b) The full systematic name for Fe(NO₃)₂ is:

- Iron(II) nitrate(V); [1 mark]

[Total: 1 mark]

- Iron has a 2+ charge, therefore its oxidation number is +2, shown by the Roman numeral (II)
- The nitrate ion has a 1- charge, so the overall oxidation number of the nitrate ion is -1
- Each oxygen atom has an oxidation number of -2 and there are 3 oxygen atoms in each nitrate ion, therefore the oxidation number due to oxygen is -6
- The sum of the oxidation numbers of nitrogen and oxygen must equal the oxidation number of the ion
 - $N + (-6) = -1$
- Therefore the oxidation number of nitrogen is +5, shown by the Roman numeral (V)
- Roman numerals for the nitrate ion are often not given in the name, however, this question specifies the full systematic name, so you must include it here to gain the mark

1c

c)

i) The species that has been reduced is:

- Co has been reduced; [1 mark]
- From +3 (in Co_2O_3) to +2 (in CoCl_2); [1 mark]

ii) Explaining if the student is correct in describing this as a disproportionation reaction:

- The reaction is not disproportionation

AND

Disproportionation is a reaction in which the same species is reduced and oxidised; [1 mark]

- Cl has been oxidised from -1 in HCl to 0 in Cl_2 ; [1 mark]
- Cl in CoCl_2 is -1 so the oxidation number remains unchanged (from -1 in HCl); [1 mark]

[Total: 5 marks]

- When explaining if a species has been reduced or oxidised, always give the oxidation numbers of both of the species it appears in and use the difference between them to explain whether it has been oxidised or reduced
- In part (ii), you need to show the examiner that you have an understanding of what

disproportionation is, so make sure you give a definition

- If you did not score the final 2 marks, you would score 1 mark if you correctly stated the oxidation number of chlorine in HCl , CoCl_2 , and Cl_2
- In this reaction, chlorine is only partially oxidised as only 2 of the 6 Cl present are oxidised

1d

d)

i) The correct oxidation numbers are:

Mg in MgO **+2**

N in HNO_3 **+5**

Mg in $\text{Mg}(\text{NO}_3)_2$ **+2**

N in $\text{Mg}(\text{NO}_3)_2$ **+5**

- All correct; [2 marks]
- 2 or 3 correct: [1 mark]

ii) Explaining whether the reaction between magnesium oxide and nitric acid is a redox reaction:

- It is not a redox reaction because there is no change in oxidation numbers
- OR**
- Mg and N are not oxidised or reduced; [1 mark]

[Total: 3 marks]

- A redox reaction involves the oxidation of one species and the reduction of another
- In this case, all of the oxidation numbers remain the same, so this is not a redox reaction

2a

a) The oxidation number of sulfur in SCl_2 is:

- +2; [1 mark]

[Total: 1 mark]

- This is a straightforward oxidation number to calculate
- Chlorine has the oxidation number -1 so for $2 \times \text{Cl} = -2$
- Overall, the oxidation number of SCl_2 is 0, therefore the oxidation number of S must be +2:
 - O.N. of S + O.N of $2 \times \text{Cl} = 0$
 - $+2 + (2 \times -1) = 0$

2b

b)

i) The changes in oxidation number of sulfur during the reaction of SCl_2 with water are:

- (Half) the sulfur goes up by 2 / +2; [1 mark]
- (The other half) goes down by 2 / -2; [1 mark]

ii) The type of reaction, with reason, is:

- Disproportionation; [1 mark]
- As sulfur is both oxidised and reduced

OR

The oxidation number of sulfur both increases and decreases; [1 mark]

[Total: 4 marks]

- The oxidation number of sulfur goes from +2 in SCl_2 to +4 in SO_2 , therefore it changes by +2 and is oxidised
- The oxidation number of sulfur goes from +2 in SCl_2 to 0 in S, therefore it changes by -2 and is reduced

2c

c)

i) The identity of compound **B** is:

- HCl; [1 mark]

ii) The equation for the reaction between SCl_2 and water is:

- $2\text{SCl}_2 + 2\text{H}_2\text{O} \rightarrow \text{S} + \text{SO}_2 + 4\text{HCl}$; [1 mark]

[Total: 2 marks]

- The question provides most of the information
- You are given the reactants: ' SCl_2 reacts with water'
- The products are 'a yellow precipitate of sulfur and a solution **A**. Solution **A** contains a mixture of $\text{SO}_2(\text{aq})$ and compound **B**', so you are given S and SO_2 as products:
 - $\text{SCl}_2 + \text{H}_2\text{O} \rightarrow \text{S} + \text{SO}_2 + \text{compound B}$
- Look to see what other elements need to be accounted for – it should be a logical conclusion that compound **B** is HCl
- The equation then needs to be balanced
- You can score the first mark if you identify HCl as compound **B** in part (ii)

2d

d)

i) When each of the reagents is added to separate samples of solution **A**, you would observe:

- With AgNO_3 : a white precipitate / ppt; [1 mark]

i) When each of the reagents is added to separate samples of solution **A**, you would observe:

- With AgNO_3 : a white precipitate / ppt; [1 mark]
- With $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$: the solution turns green; [1 mark]

ii) The systematic name of KMnO_4 is:

- Potassium manganate(VII); [1 mark]

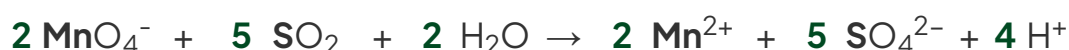
[Total: 3 marks]

- Silver nitrate will react with the chloride ions to form the white precipitate, AgCl
 - $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$
- Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ is an oxidising agent
- When it is added to solution **A**, the orange dichromate ions ($\text{Cr}_2\text{O}_7^{2-}$) are reduced to green Cr^{3+} ions:
- The oxidation number of manganese in MnO_4^- is +7, which is shown by the Roman numeral (VII)
 - O.N. due to oxygen is $4 \times -2 = -8$
 - O.N. of MnO_4^- is -1
 - O.N. of Mn + (-8) = -1
 - Therefore O.N. of Mn must be +7

3a

a)

i) The correct oxidation numbers and balanced equation is:



oxidation
numbers:

+7

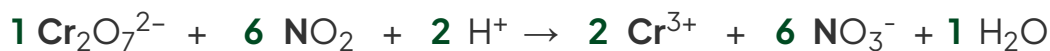
+4

+2

+6

- Oxidation numbers; [1 mark]
- Balancing of redox species; [1 mark]
- Balancing H_2O and H^+ ; [1 mark]

ii) The correct oxidation numbers and balanced equation is:



oxidation

numbers:

+6

+4

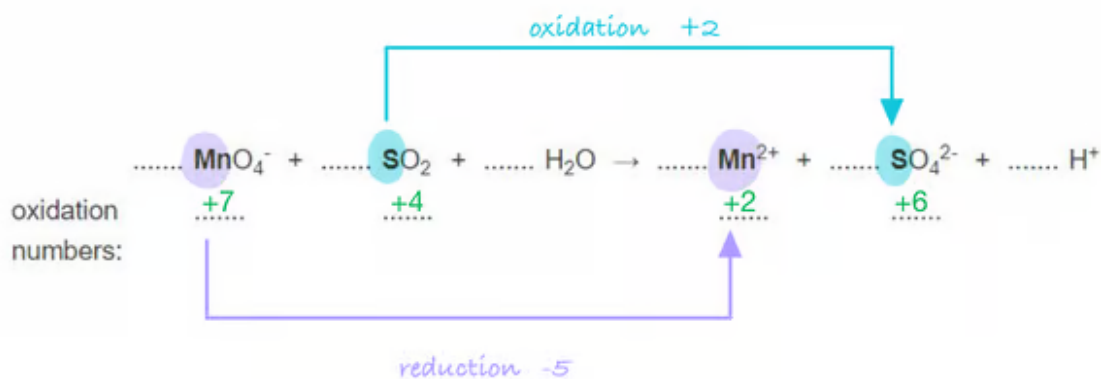
+3

+5

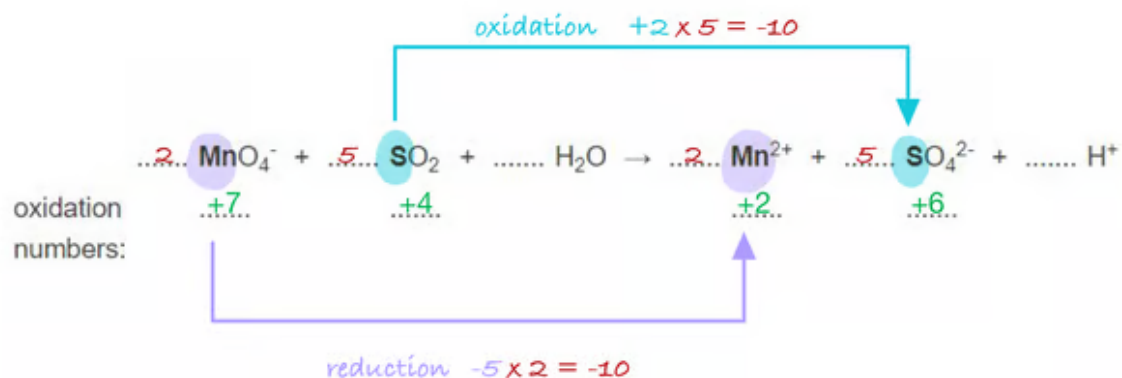
- Oxidation numbers; [1 mark]
- Balancing of redox species; [1 mark]
- Balancing H_2O and H^+ ; [1 mark]

[Total: 6 marks]

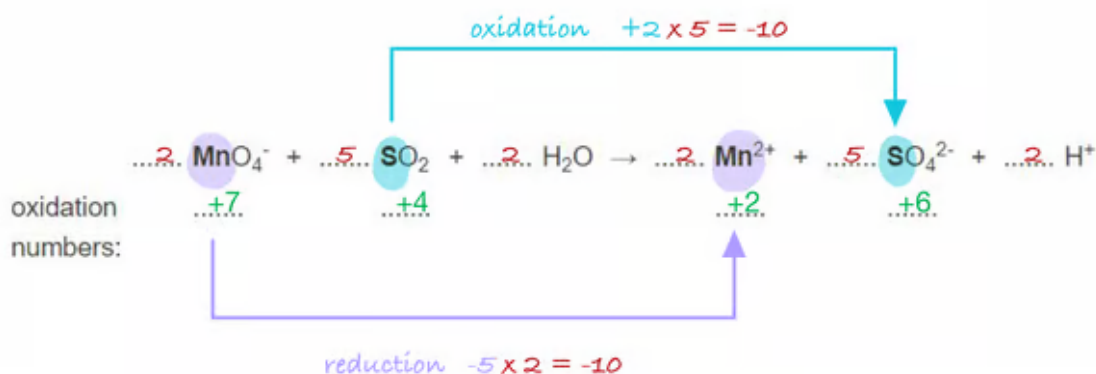
- You must use oxidation numbers to balance the equation, which can be done using the following steps:
- Once you have deduced the oxidation numbers, balance the redox species (in this case they are already balanced) and then identify the changes in oxidation numbers:



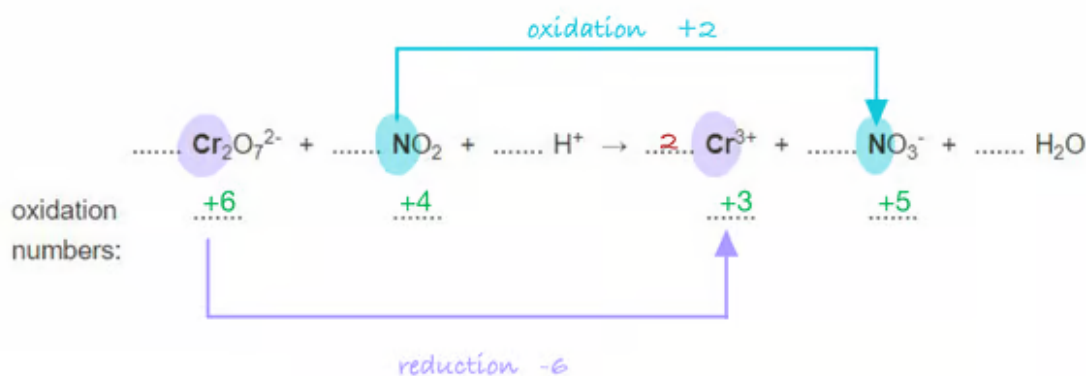
- Balance the changes in oxidation number so that they are equal (but opposite in sign) which determines the balancing numbers of the redox species:



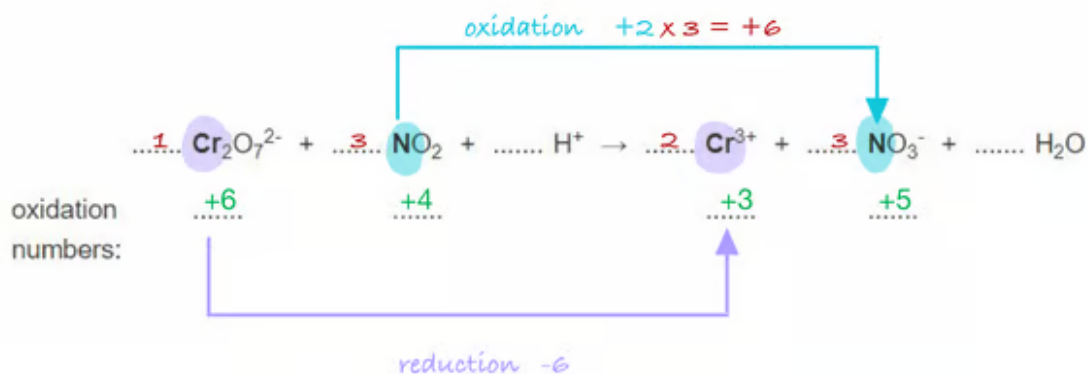
- Finally, balance the oxygen atoms by adding an appropriate balancing number in front of H_2O , then balance the hydrogen atoms by adding an appropriate balancing number in front of H^+ :



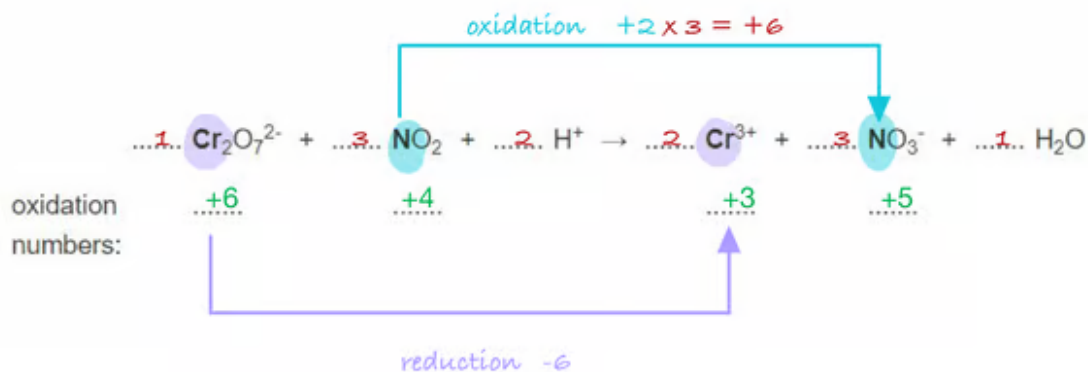
- The same can be done for the second equation
- Once you have deduced the oxidation numbers, balance the redox species (in this case there are 2 Cr) and then identify the changes in oxidation numbers – remember that each Cr will decrease by -3, so 2 x Cr will decrease by -6:



- Balance the changes in oxidation number so that they are equal (but opposite in sign) which determines the balancing numbers of the redox species:



- Finally, balance the oxygen atoms by adding an appropriate balancing number in front of H_2O , then balance the hydrogen atoms by adding an appropriate balancing number in front of H^+ :



3b

b) What you would you **see** happening in the reaction shown in **(a)(ii)** is:

- (Turns from) orange; [1 mark]
- To green; [1 mark]

[Total: 2 marks]

- The dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$, is an orange coloured oxidising agent
- In this reaction, the dichromate ion is itself reduced to Cr^{3+} , which are green coloured ions
- The use of the dichromate ions as an oxidising agent and the resulting colour change should be familiar to you from the oxidation of alcohols

3c

c)

i) The balanced symbol equation for the reaction is:

- $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$; [1 mark]

ii) Explaining, using oxidation numbers, why this is a redox reaction:

- The oxidation number of Zn goes from 0 (in Zn) to +2 (in ZnSO_4) so is oxidised; [1 mark]
- The oxidation number of Cu goes from +2 (in CuSO_4) to 0 (in Cu) so is reduced; [1 mark]

[Total: 3 marks]

- The equation should be straightforward, the products will just have the Zn and Cu swapped around as Zn displaces Cu
- When explaining whether a species is oxidised or reduced, makes sure you give the oxidation number before and after the reaction has taken place
- An increase in oxidation number means the species has been oxidised
- A decrease in oxidation number means the species has been reduced

3d

d) Using a half equation to explain what species is acting as the oxidising reagent in the

ii) Explaining, using oxidation numbers, why this is a redox reaction:

- The oxidation number of Zn goes from 0 (in Zn) to +2 (in ZnSO_4) so is oxidised; [1 mark]
- The oxidation number of Cu goes from +2 (in CuSO_4) to 0 (in Cu) so is reduced; [1 mark]

[Total: 3 marks]

- The equation should be straightforward, the products will just have the Zn and Cu swapped around as Zn displaces Cu
- When explaining whether a species is oxidised or reduced, makes sure you give the oxidation number before and after the reaction has taken place
- An increase in oxidation number means the species has been oxidised
- A decrease in oxidation number means the species has been reduced

3d

d) Using a half equation to explain what species is acting as the oxidising reagent in the reaction in (c):

- $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$; [1 mark]
- Cu^{2+} gains electrons so is reduced; [1 mark]
- Which causes Zn to lose electrons so is oxidised; [1 mark]

[Total: 3 marks]

- The number of marks should indicate the level of detail required in this response
- Your team technique should tell you that there is 1 mark available for the half equation, therefore 2 marks for the explanation
- The use of the half equation should prompt you to explain this in terms of electrons

4a

a)

i) The conditions required for NaClO to be produced from chlorine and sodium hydroxide are:

- Cold / 15 °C; [1 mark]

ii) The systematic name of NaClO is:

- Sodium chlorate(I); [1 mark]

iii) The formula for sodium chlorate(V) is

- NaClO₃; [1 mark]

[Total: 3 marks]

- When cold dilute alkali is reacted with chlorine, NaClO, H₂O and NaCl are formed
- When hot concentrated alkali is reacted with chlorine, NaClO₃, H₂O and NaCl are formed
- Roman numerals show the oxidation state of chlorine
 - In NaClO: It is +1 as Na = +1, O = -2 and overall = 0
 - In NaClO₃ it is +5 as Na = +1, O = 3 x -2 = -6 and overall = 0

4b

b)

i) The ionic equation for this reaction is:

- $\text{Cl}_2 + 2\text{OH}^- \rightarrow \text{Cl}^- + \text{ClO}^- + \text{H}_2\text{O}$; [1 mark]

ii) This is an example of a disproportionation reaction because:

- (Disproportionation is a reaction in which) the same species is both oxidised and reduced; [1 mark]
- Cl is oxidised from 0 (in Cl₂) to +1 (in ClO⁻); [1 mark]
- Cl is reduced from 0 (in Cl₂) to -1 (in NaCl); [1 mark]

[Total: 4 marks]

- When writing the ionic equation, rewrite the equation substituting in any ions present:
 - $\text{Cl}_2 + 2\text{Na}^+ + 2\text{OH}^- \rightarrow \text{Na}^+ + \text{Cl}^- + \text{Na}^+ + \text{ClO}^- + \text{H}_2\text{O}$
- Cancel out any spectator ions to give the ionic equation



- In part (ii), you need to demonstrate that you understand the meaning of disproportionation

4c

c) Explaining, using oxidation numbers, why chlorine has been reduced in this reaction:

- The oxidation number of Cl has decreased from +1 (in NaClO) to -1 (in NaCl) (so has been reduced); [1 mark]

[Total: 1 mark]

- To show oxidation or reduction using oxidation numbers, you must show the oxidation number of the element before and after the reaction and state whether the oxidation number has increased or decreased
- **Remember:** An increase in oxidation number shows oxidation
- A decrease in oxidation shows reduction

4d

d)

i) The number of moles of oxygen gas that were collected is:

- $n(\text{O}_2) = 108 \div 24\,000 = 4.5 \times 10^{-3} \text{ (mol)}$

OR

$$0.108 \div 24 = 4.5 \times 10^{-3} \text{ (mol)}; [1 \text{ mark}]$$

ii) The number of moles of NaClO gas that were in the 10.0 cm^3 sample of bleach is:

- $n(\text{NaClO}) = n(\text{O}_2) = 4.5 \times 10^{-3} \text{ (mol)}; [1 \text{ mark}]$

iii) The concentration of NaClO, in mol dm^{-3} , of NaClO in the sample of bleach is:

- Concentration = $\frac{4.5 \times 10^{-3}}{10.0} \times 1000 = 0.45 \text{ mol dm}^{-3} \text{ (mol dm}^{-3}\text{)}; [1 \text{ mark}]$

iv) The concentration of NaClO in g per cm^3 is:

- Convert mol dm^{-3} to g dm^{-3} = $0.45 \times 74.5 = 33.525 / 33.5 \text{ g dm}^{-3}$; [1 mark]
- Convert to g per 100 cm^3 = $\frac{33.525}{1000} \times 100 = 3.3525 / 3.35 \text{ (g per } 100 \text{ cm}^3)$; [1 mark]

[Total: 5 marks]

- When calculating the volume of oxygen gas, ensure that you either convert the volume of oxygen to dm^3 before using the molar gas volume of 24.0, or use moles = volume (cm^3) $\div$ 24 000
- The stoichiometry of the equation in part c) tells you that 1 mol of oxygen is produced from 1 mole of NaClO
- When calculating the concentration in mol dm^{-3} , remember to convert the volume to dm^3
- g per 100 cm^3 is not a common unit of concentration
- First, find the concentration in g dm^{-3} by multiplying the concentration in mol dm^{-3} by the M_r of NaClO
 - $M_r = 23.0 + 35.5 + 16 = 74.5$
 - Concentration = $0.45 \times 74.5 = 33.525 \text{ g dm}^{-3}$
- So there are 33.525 g in every 1 dm^3 , which is 1000 cm^3
- So in 100 cm^3 there will be 10 x times less than this, so 3.3525 g per 100 cm^3

5a

a)

i) The balanced equation for this reaction is:

- $\text{Cl}_2 + 2\text{KI} \rightarrow 2\text{KCl} + \text{I}_2$; [1 mark]

ii) The ionic equation for this reaction is:

- $\text{Cl}_2 + 2\text{I}^- \rightarrow 2\text{Cl}^- + \text{I}_2$; [1 mark]

[Total: 2 marks]

- These should be easy marks to pick up as this is a reaction you should be very familiar with

- When writing the ionic equation, the spectator ion, K^+ , will cancel out and remember to include the balancing number of the halide ions

5b

b) Explaining how iodide ions act as a reducing agent in this reaction:

- Reducing agents decrease the oxidation number of another atom; [1 mark]
- Iodide ions / I^- cause the oxidation number of Cl to decrease from 0 (in Cl_2) to -1 (in KCl); [1 mark]

OR

- The reducing agent is the substance that gets oxidised; [1 mark]
- The oxidation number of I increases from -1 (in I^-) to 0 (in I_2) so I is oxidised; [1 mark]

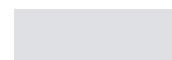
[Total: 2 marks]

- You need to explain what a reducing agent
- **Careful:** The question asks you to explain in terms of oxidation number, so any reference to electrons will be ignored

5c

c)

i) You can use these results to determine which is the stronger reducing agent, bromide ions or iodide ions as:



- You need to explain what a reducing agent
- **Careful:** The question asks you to explain in terms of oxidation number, so any reference to electrons will be ignored

5c

c)

i) You can use these results to determine which is the stronger reducing agent, bromide ions or iodide ions as:

- A stronger reducing agent will donate electrons more easily; [1 mark]
- Iodide ions donate electrons more easily (than bromide ions) to chlorine so are the stronger reducing agent; [1 mark]

ii) To explain why the halide ions show this trend in reducing power:

- Going down the group, the halide ions become larger / ionic radius increases; [1 mark]
- The outermost electrons get further away from the nucleus

OR

The outer electrons experience more shielding (by inner electrons); [1 mark]

- There is less attraction between the outer electrons and nucleus; [1 mark]
- (So) the halide ions lose electrons more easily going down the group (increasing their reducing power); [1 mark]

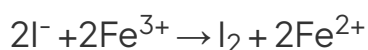
[Total: 6 marks]

- The reactions of the halogens and halides are commonly used to demonstrate various redox reactions
- The explanation for the trend in reducing power is worth learning

5d

d)

i) The ionic equation for this reaction is:



- Correct reactants and products; [1 mark]
- Correct balancing; [1 mark]

ii) Fe^{3+} ions act as an oxidising agent because:

- Oxidising agents accept electrons from another species / atom; [1 mark]
- Fe^{3+} accepts an electron from I^- ; [1 mark]

[Total: 4 marks]

- To construct the ionic equation, you can think about the two half-equations
- Iodide is oxidised as it loses electrons:
 - $2\text{I}^- \rightarrow \text{I}_2 + 2\text{e}^-$
- Fe^{3+} is reduced as it gains electrons:
 - $\text{Fe}^{3+} + \text{e}^- \rightarrow \text{Fe}^{2+}$
- To combine the two half equations, multiply the iron equation by two, so that both half equations contain 2e^- :
 - $2\text{Fe}^{3+} + 2\text{e}^- \rightarrow 2\text{Fe}^{2+}$
- Add all of the reactants from both half equations and all of the products – the 2e^- will cancel out to give the ionic equation
 - $2\text{I}^- + 2\text{Fe}^{3+} \rightarrow \text{I}_2 + 2\text{Fe}^{2+}$
- $\text{I}^- + \text{Fe}^{3+} \rightarrow \frac{1}{2}\text{I}_2 + \text{Fe}^{2+}$ would have also gained the marks
- This explanation of why Fe^{3+} acts as an oxidising agent requires you to explain in terms of electrons, so any reference to oxidation numbers is ignored

